

Photovoltaic - MC4 String Connector: Field Installation, Crimp & Inspection Specification

1. Scope and applicable standards

This specification governs the field termination, mating, inspection and maintenance of MC4-type photovoltaic (PV) string connectors used on module and string DC wiring. It applies to new installation and to in-service repair / re-termination.

- IEC 62852 - connectors for DC application in PV systems (safety + intermateability)
- UL 6703 - connectors for use in PV systems
- IEC 62548 - PV array design requirements
- IEC 62446 - PV systems: commissioning, documentation and inspection

2. Connector ratings (typical)

Parameter Typical value

- Rated voltage | 1000 / 1500 V DC
- Rated current | 30 A (4 mm²) up to ~45 A (10 mm²)
- Temperature range | -40 deg C to +85/+90 deg C
- Ingress protection (mated) | IP67 / IP68
- Contact resistance (new) | < 0.5 mOhm
- Retention (pull-out) force | several hundred N per type

3. Tools and materials

- Matched MC4 crimp tool with the correct die for the wire cross-section
- PV cable stripper sized to the conductor
- Pair of connector spanners / torque spanner for the gland
- GENUINE connectors of a single make matched to the cable (see cross-mating warning)

4. Cable compatibility

Cable CSA ~AWG Notes

- 4 mm² | 12 | common residential strings
- 6 mm² | 10 | common C&I strings
- 10 mm² | 8 | long strings / high current

Use double-insulated, tinned-copper PV wire (e.g. H1Z2Z2-K). Strip length is per the connector datasheet, typically 6-7 mm. Do not use untinned building wire.

5. Crimp procedure [crimp]

1. Strip the conductor to the specified length without nicking strands.
2. Insert ALL strands fully into the contact barrel - no strand left outside.
3. Crimp with the matched tool and die to the specified crimp height.
4. Verify: every strand captured, no splay, correct crimp height, no cut strands.
5. Gently pull-test the contact - it must not move in the barrel.
6. Seat the contact in the housing until the audible click; torque the gland to datasheet (typ. 2-3 Nm).

6. Crimp quality - acceptance and rejection

Defect Cause Action

- Stray / splayed strands | strands not fully inserted | re-strip and re-crimp
- Loose (under-crimp) | wrong die / low force | reject, re-crimp

- Cut strands (over-crimp) | die too small / over-force | reject, re-terminate
- Oxidised copper | old or exposed wire | cut back to bright copper

7. Contact resistance and the overheating physics

A new, correctly crimped and fully mated contact is below 0.5 mOhm. A degraded joint (loose crimp, oxidation, cross-mating) can climb to tens of mOhm. Heat generated at the joint is I^2R . Worked example: at 10 A string current a 50 mOhm joint dissipates $I^2R = 10^2 \times 0.050 = 5 \text{ W}$ in a tiny contact area. That local heat softens and melts the housing and oxidises the contact further, which raises the resistance again - a thermal runaway that ends in charring and a self-sustaining DC series arc. Frayed or splayed copper visible at the crimp is a classic indicator of the defective termination that starts this chain.

8. Mating and environmental sealing

- Mate only GENUINE, same-type, same-make halves until the latch clicks.
- The gland and O-ring provide the IP67/IP68 seal; an under-torqued gland admits water, which oxidises the contact and raises resistance.
- Never re-use a connector that shows any heat discolouration or melt.

9. Inspection and preventive maintenance

- Commission per IEC 62446 (continuity, insulation, polarity, Voc/Isc).
- Periodically IR-thermograph connectors UNDER LOAD - a connector hotter than its neighbours signals a high-resistance joint before it fails.
- Re-torque or replace flagged connectors; record make, part number and date.